Test II

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Novel Prep

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Phase 3 · Test II

Overview

Test II

22 word-problem training challenges

Question 1

A computer program models the total mass of the population of a certain type of algae after the algae was placed in an environment where it has no natural predators. According to the model, the estimated total mass of this population of algae at the end of every 6-hour period is 129% greater than the estimated total mass of this population of algae at the end of the previous 6-hour period, and the estimated total mass of this population of algae is 613.90 grams after 18 hours. Which equation best represents this model, where A is the estimated total mass, in grams, of the population of algae after x hours, and $x < 50$?

A. $A = 34.11(1.29)^{\frac{x}{6}}$

B. $A = 34.11(2.29)^{\frac{x}{6}}$

C. $A = 51.12(2.29)^{\frac{x}{6}}$

D. $A = 285.98(1.29)^{\frac{x}{6}}$

Answer 1

Correct Answer: C

Question 2

A solar panel is supported by two braces that form acute angles X and Y with two perpendicular sides of its frame. The measure of angle Y is 32°, and a load calculation for the braces gives $\dfrac{\sin X}{\cos Y} = 1$. What is the measure, in degrees, of angle X?

A. 77

B. 58

C. 45

D. 32

Answer 2

Correct Answer: B

Question 3

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(7x - 410)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-14x + w)^\circ$. What is the value of w?

Answer 3

Final Answer: $\boxed{1360}$

Question 4

The equation $M = 1{,}800(1.02)^t$ models the number of members, M, of a gym t years after the gym opens. Of the following, which equation models the number of members of the gym q quarter years after the gym opens?

A. $M = 1{,}800(1.02)^{\frac{q}{4}}$

B. $M = 1{,}800(1.02)^{4q}$

C. $M = 1{,}800(1.005)^{4q}$

D. $M = 1{,}800(1.082)^{q}$

Answer 4

Correct Answer: A

Question 5

An engineer uses the equation

$$\frac{1}{24}x^2 + \left(s - \frac{1}{24}t\right)x - st = 0$$

to determine two possible signed calibration adjustments, x, for a machine. In the equation, s and t are positive constants. The product of the two possible adjustments is $-2kst$, where k is a constant. What is the value of k?

Answer 5

Final Answer: $\boxed{12}$

Question 6

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y, in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 1.75 inches, it has an estimated breaking strength of 3,858.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 3.50 inches?

Answer 6

Final Answer: $\boxed{15435}$

Question 7

A hollow steel pipe is in the shape of a right circular cylinder. The steel pipe has an outside diameter of 72 inches, a wall thickness of $\frac{5}{8}$ inches, and a height of 138 inches. Which of the following is closest to the volume, in cubic inches, of the wall of this steel pipe?

A. 169

B. 542

C. 3,105

D. 19,340

Answer 7

Correct Answer: D

Question 8

Trapezoid $CDEF$ is similar to trapezoid $JKLM$, such that C, D, E, and F correspond to J, K, L, and M, respectively. The length of each side of trapezoid $JKLM$ is 4 times the length of its corresponding side in trapezoid $CDEF$. The measure of angle C is 42°. What is the measure of angle J?

A. 38°

B. 42°

C. 46°

D. 168°

Answer 8

Correct Answer: B

Question 9

A robotic arm rotates through an angle F of $\frac{\pi}{15}$ radians. Its control panel displays this rotation as $6n$ degrees, where n is a constant. What is the value of n?

Answer 9

Final Answer: $\boxed{2}$

Question 10

The table gives the average time t, in minutes, it takes Adriana to travel a certain distance d, in miles. Which equation could represent this linear relationship?

Distance (miles)	Average time (minutes)
0.18	9
0.24	12
0.36	18

A. $t = 50d$

B. $t = \frac{1}{2}d$

C. $t = 2d$

D. $t = \frac{1}{50}d$

Answer 10

Correct Answer: A

Question 11

A wireless phone company pays customers for trading in old smartphones. The payment p, in dollars, is the original retail price of a smartphone minus \$135 for each year since the smartphone was released. Which equation represents this situation for a smartphone with an original retail price of \$750 when it was released t years ago, where t is a whole number and $1 \leq t \leq 4$?

A. $p = 135 - 750t$

B. $p = 750 - 135t$

C. $t = 135 - 750p$

D. $t = 750 - 135p$

Answer 11

Correct Answer: B

Question 12

A soccer team's goal is to earn at least \$1,170 by selling coupon books. The team earns \$9 from selling each coupon book. Which of the following inequalities describes all possible values for the number of coupon books, n, the team can sell to meet the goal?

A. $n + 9 \leq 1{,}170$

B. $n + 9 \geq 1{,}170$

C. $9n \leq 1{,}170$

D. $9n \geq 1{,}170$

Answer 12

Correct Answer: D

Question 13

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 62 sedans for sale at the dealership on Monday, how many minivans were for sale?

Answer 13

Final Answer: $\boxed{93}$

Question 14

A snack company advertises that bags of pretzels produced by the company weigh, on average, 1 pound each. To test this, Sam selected at random 30 bags of pretzels produced by the company and weighed each bag. Based on his measurements, Sam estimated that for all bags of pretzels produced by the company, the average weight per bag is 1.02 pounds, with a margin of error of 0.05 pounds. For all bags of pretzels produced by the company, which of the following is the most appropriate conclusion about the average weight w, in pounds, per bag?

A. $w = 1.02$

B. $0.97 \leq w \leq 1.02$

C. $0.97 \leq w \leq 1.07$

D. $1.02 \leq w \leq 1.07$

Answer 14

Correct Answer: C

Question 15

A total of 3 equilateral pentagons each have side length p units. A total of 2 squares each have side length s units. The sum of the perimeters of all these pentagons and squares is 136 units. Which equation represents this situation?

A. $12p + 10s = 136$

B. $2p + 3s = 136$

C. $3p + 2s = 136$

D. $15p + 8s = 136$

Answer 15

Correct Answer: D

Question 16

A company conducted a study concerning the time spent on extracurricular activities by students in grades 6, 7, and 8 and their level of school spirit. A total of 80 students were selected at random from all students in grades 6, 7, and 8 in a large school district in the US. The selected students were asked to report their level of school spirit as either low, medium, or high. The table summarizes the distribution of the selected students by grade.

Grade	6	7	8
Number of students	26	25	29

If the results of the study suggest a relationship between the self-reported level of school spirit and the amount of time spent on extracurricular activities by students in grades 6, 7, and 8, which of the following aspects of the study would prevent the results from being generalized to all students in grades 6, 7, and 8 in the US?

A. The selected students are not all from the same school.

B. The number of selected students is too small.

C. The number of selected students from each grade is different.

D. The students were not selected at random from all students in grades 6, 7,

Answer 16

Correct Answer: D

Question 17

For a study, a research team measured the heights of 240 king penguins from the Crozet Islands and 120 king penguins from the Falkland Islands. The research team determined that the mean height of the 240 king penguins from the Crozet Islands was 86 centimeters and the mean height of the 120 king penguins from the Falkland Islands was 92 centimeters. What was the mean height, in centimeters, of all 360 king penguins the research team measured for this study?

A. 87

B. 88

C. 89

D. 90

Answer 17

Correct Answer: B

Question 18

At a nature preserve, a wildlife biologist counted bald eagles from an observation deck at the same time each day for 21 days. The table summarizes the resulting data set, data set A.

Number of bald eagles	Number of days
0	1
1	3
2	4
3	5
4	4
5	3
19	1

The data value 19 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 20 data values. Which statement best compares the median of data set A and the median of data set B?

A. The median of data set B is less than the median of data set A.

B. The median of data set B is equal to the median of data set A.

C. The median of data set B is greater than the median of data set A.

Answer 18

Correct Answer: B

Question 19

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B.

Right circular cylinder A	392π
Right circular cylinder B	$10{,}584\pi$

The radius of right circular cylinder A is 7 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

A. 1008

B. 1210

C. 1680

D. 2014

Answer 19

Correct Answer: C

Question 20

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Bennett opened one of these accounts with an original account balance of \$700 and did not make any other deposits or withdrawals. 4 months after Bennett opened the account, the balance had increased by 0.6% of the original balance. 6 months after Bennett opened the account, the balance had increased by an additional 0.2% of the balance at the end of the first 4 months. Every 2 months after the first 6 months, the balance had increased by an additional 0.2% of the balance 2 months before. Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 4$?

A. $B(x) = 704.20(1 + 0.002)^{\frac{x}{2} - \frac{4}{2}}$

B. $B(x) = 704.20(1 + 0.002)^{\frac{x}{2} - 4}$

C. $B(x) = 704.20(1 + 0.002)^{2x - 8}$

D. $B(x) = 704.20(1 + 0.002)^{2x - 4}$

Answer 20

Correct Answer: A

Question 21

In an architecture elective, students study a triangular support panel used on a small footbridge. On the scale drawing, the panel is labeled as triangle JKL, with vertex J at the top joint and vertices K and L along the base of the panel. The teacher gives the class the angle measures from the drawing: the measure of angle J is $90b^\circ$, the measure of angle K is $66a^\circ$, and the measure of angle L is $24a^\circ$, where a and b are constants. Before the students calculate any side lengths for a load estimate, they are asked which comparison of $\cos L$ and $\sin K$ must be true based only on this information.

A. $\cos L > \sin K$

B. $\cos L = \sin K$

C. $\cos L < \sin K$

D. There is not enough information to compare the values of $\cos L$ and $\sin K$.

Answer 21

Correct Answer: D

Question 22

A chemistry lab uses two related formulas when converting a sensor reading into a calibrated output. For a positive constant p that depends on the sensor model, the intermediate reading is given by the function j defined by $j(x) = \dfrac{4x+2}{9p}$, and the calibrated output is given by the function k defined by $k(x) = \dfrac{j(x) \cdot p}{x+1}$. During one trial, the lab records an input value of $x = 3$. What is the value of $k(3)$?

A. $\dfrac{7}{18}$

B. $\dfrac{4}{9}$

C. $\dfrac{14}{13}$

D. $\dfrac{10}{9}$

Answer 22

Correct Answer: A

Thank You!

We appreciate your time and focus.

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